

4

BIOLUMINESCENCE

Production and emission of light by living organisms

Bioluminescence is the production and emission of light by living organisms. It is a form of chemiluminescence where light energy is released by a chemical reaction within an organism. Bioluminescence occurs widely in marine vertebrates and invertebrates, as well as in some fungi, microorganisms including some bioluminescent bacteria, and terrestrial arthropods such as fireflies.

Bioluminescence is the production and emission of light by living organisms. It is a form of chemiluminescence where light energy is released by a chemical reaction within an organism. Bioluminescence occurs widely in marine vertebrates and invertebrates, as well as in some fungi, microorganisms including some bioluminescent bacteria, and terrestrial arthropods such as fireflies. The phenomenon occurs through a chemical reaction involving the light-emitting molecule luciferin and the enzyme luciferase. When luciferin reacts with oxygen in the presence of luciferase, light is produced. Different organisms have evolved different forms of luciferin and luciferase independently. Marine bioluminescence is particularly widespread. An estimated 76 percent of ocean animals produce light. In the deep sea, where sunlight never penetrates, bioluminescence is the primary source of light. Many deep-sea creatures use bioluminescence for communication, camouflage, or hunting. Fireflies use bioluminescence for mating displays. Male fireflies fly through the air flashing specific patterns to attract females. Different firefly species have unique flash patterns, serving as a species-recognition system that prevents interbreeding. Some organisms use bioluminescence for defense. When disturbed, certain squid release bioluminescent ink to confuse predators. Deep-sea creatures may use sudden flashes to startle attackers or to make

Bioluminescent bacteria form symbiotic relationships with various marine animals. The Hawaiian bobtail squid hosts luminescent bacteria in special light organs. The squid uses this light for counter-illumination camouflage, matching the moonlight from above to hide from predators below. Scientific research has harnessed bioluminescence for numerous applications. Green fluorescent protein (GFP) from jellyfish revolutionized cell biology, allowing scientists to track cellular processes in real-time. The discovery earned the 2008 Nobel Prize in Chemistry. Medical diagnostics increasingly use bioluminescent markers to detect diseases and track biological processes. Researchers can observe tumor growth, infection spread, and drug effectiveness in living subjects using bioluminescent imaging technologies.